Cloud Native Data Engineering

Week 1. Introduction to Cloud Data Engineering

Ashish Pujari | apujari@uchicago.edu



Lecture Outline

1. Data Engineering Foundations
2. Data Characteristics, Formats & Pipelines
3. Cloud Computing Fundamentals
4. Cloud Computing Trends
5. Cloud Success Stories



Data Engineering Foundations

Module 1

The Data Engineer Role

- Design, build, and operate reliable data systems at scale
- Bridge raw data sources and downstream consumers (analysts, scientists, apps)
- Own the data infrastructure: pipelines, storage, quality, and observability
- Key skills: SQL, Python, cloud platforms, distributed systems, DevOps
- Distinct from Data Scientist (models) and Data Analyst (insights)

DE vs Data Scientist vs Data Analyst

Data Engineer

Builds and maintains the data platform and pipelines

SQL · Python · Cloud · DevOps

Data Scientist

Trains models and extracts statistical insights from data

Stats · ML · Experimentation

Data Analyst

Queries curated data to answer business questions

SQL · BI tools · Dashboards

All three roles depend on well-engineered, trustworthy data pipelines

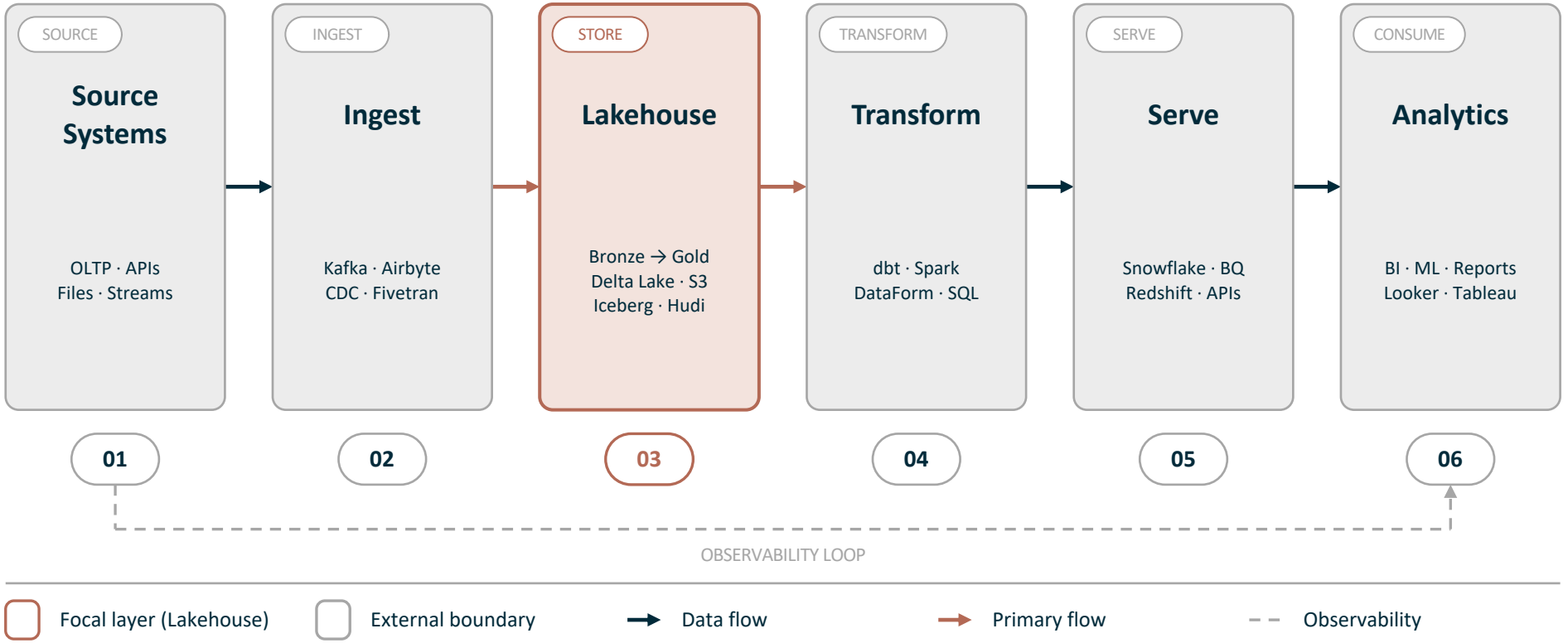
Modern teams blur boundaries — engineers often collaborate across all three

The Modern Data Stack



Data Engineering Lifecycle

Six layers every data engineer must master



Fundamentals · Ingestion · Storage · Transformation · Serving · Consumption



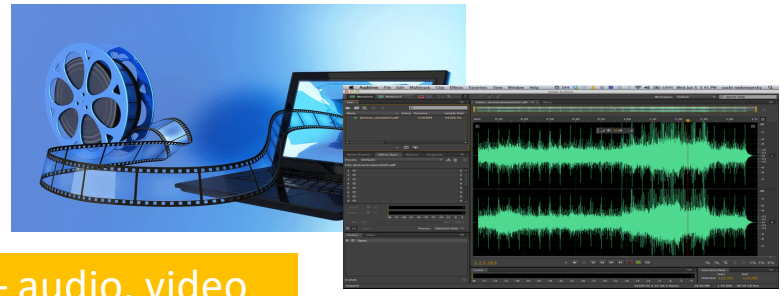
Data Characteristics, Formats & Pipelines

Module 2

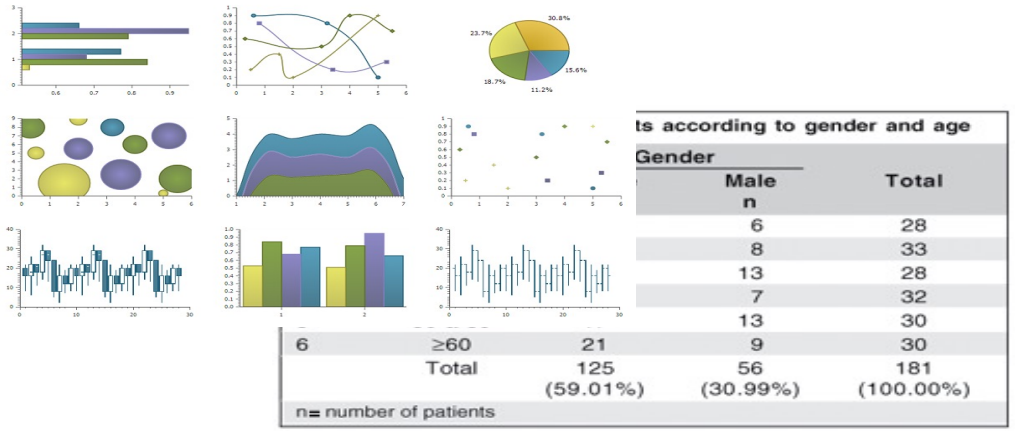
Variety of Data



Text



Media – audio, video



ts according to gender and age

Gender				
		Male n		Total
		6		28
		8		33
		13		28
		7		32
		13		30
6	≥60	21	9	30
	Total	125 (59.01%)	56 (30.99%)	181 (100.00%)

n= number of patients

Numerical



Geospatial

Data – The 4 Vs

Volume

Terabytes to petabytes — scale drives architectural choices

Velocity

Batch (hourly/daily) vs. near-real-time vs. streaming (ms)

Variety

Structured tables, JSON/XML, images, audio, video, logs

Veracity

Data quality, missing values, schema drift, source reliability

A fifth V: Value — only data that drives decisions is worth storing

Text files

- Field delimiters separate data fields. Record delimiters separate groups of fields.
- Examples
  - Comma Separated Values (,)
  - Pipe Separated Values (|)
  - Tab Separated Values (tab)
  - Semicolon separated values (;)

```
fname,lname,age,salary
nancy,davolio,33,$30000
erin,borakova,28,$25250
tony,raphael,35,$28700
```

CSV file

https://en.wikipedia.org/wiki/Delimiter/

JavaScript Object Notation (JSON)

- Lightweight data-interchange format
- Platform independent
- Built on two structures :
  - Collection of name/value pairs
  - An ordered list of values

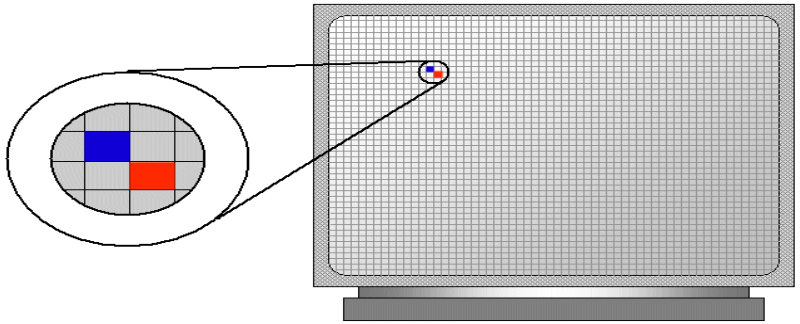
```
{
  "message": {
  "date": "07-04-2016",
  "hour": "08:30",
  "to": "Receiver",
  "from": "Sender",
  "body": "This is an email message"
}}
```

Extensible Markup Language (XML)

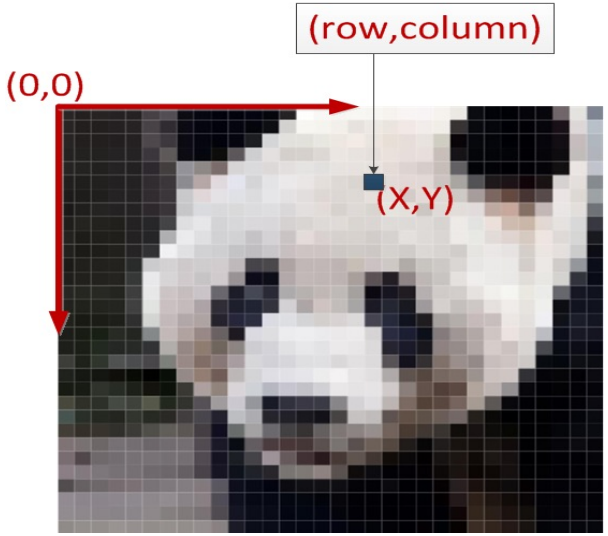
- Lightweight data-interchange format
- Platform independent
- Self-descriptive
- Extensible

```
<note>
  <date>07-04-2016</date>
  <hour>08:30</hour>
  <to>Receiver</to>
  <from>Sender</from>
  <body>This is an email message</body>
</note>
```

Digital Images



0	1	0	0	0	1	0
1	1	0	1	1	0	0
0	1	1	0	0	0	0
0	1	1	1	1	0	0
1	1	0	1	0	1	0
0	0	0	0	0	0	1

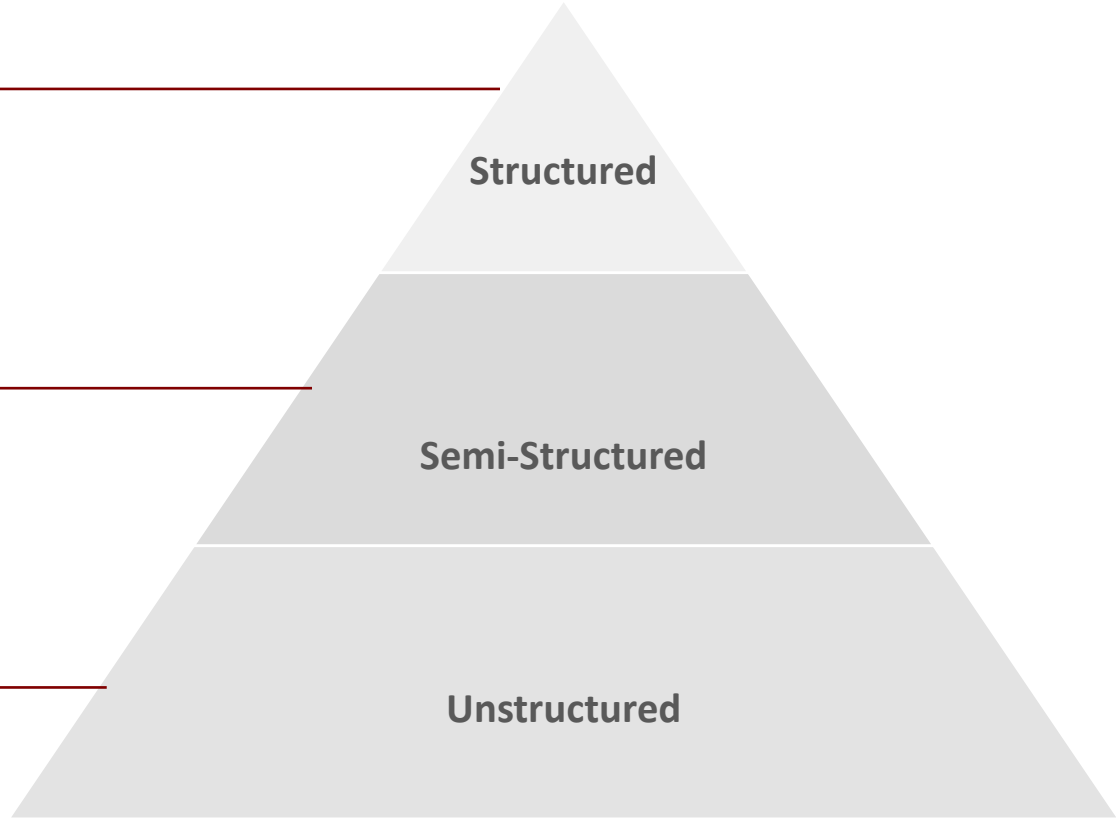


Data Types

Well defined data types,fixed fields e.g. numeric, currency, date, address, text

Textual data with patterns e.g. XML, EDI, logs

No inherent structure. File formats e.g. audio, video, pdf



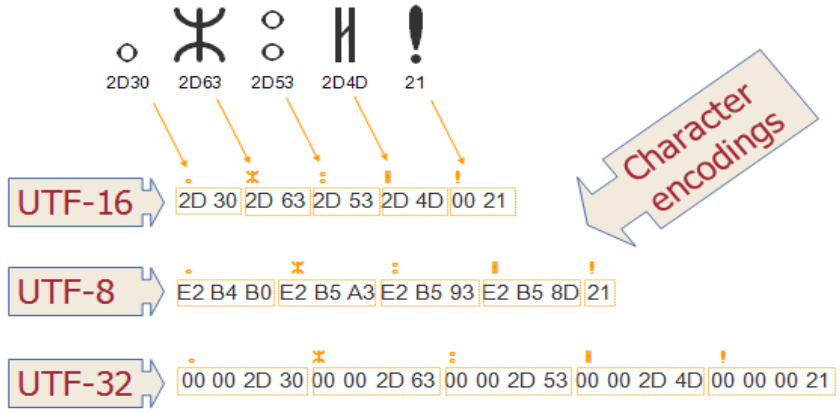
Data Formats

Category	Examples	Best for
Row formats	CSV, TSV, JSON	Human-readable, simple — slow for analytics
Columnar formats	Parquet, ORC	Analytical queries — compressed, optimized
Binary formats	Avro, Protocol Buffers	Compact, schema-enforced, streaming-friendly
Open table formats	Delta Lake, Iceberg, Hudi	ACID transactions + versioning

Format choice affects storage cost, query speed, and interoperability

Character Sets and Encodings

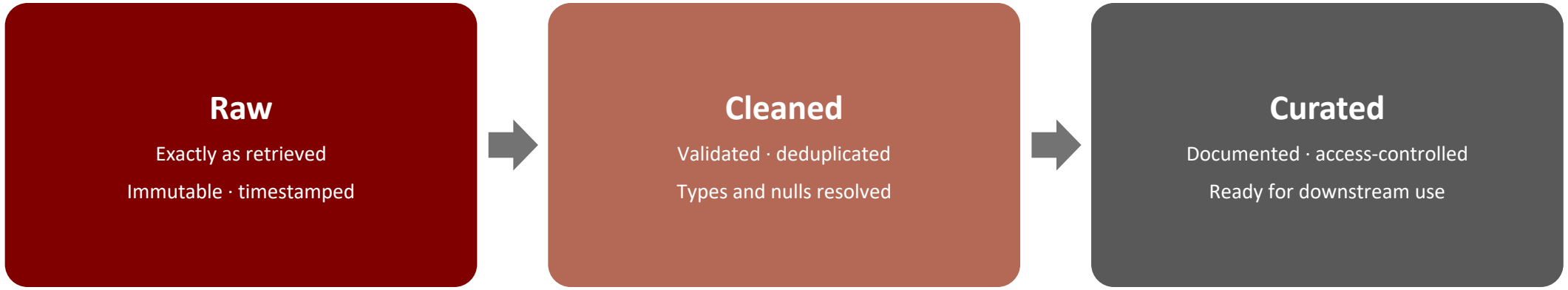
- Character Set is a collection of supported characters
- Encoding maps a string of characters to a string of bytes
- Always specify encoding when reading text files to avoid mojibake errors



Source: w3c.org

Data	Encoding	Bits
A	ASCII /UTF-8	01000001
A	UTF-16	00000000 01000001
a	ASCII /UTF-8	01100001
7	ASCII /UTF-8	00000111
bits	UTF-8	01100010 01101001 01110100 01110011
あ	UTF-8	11100011 10000001 10000010 00001010
ऋ	UTF-8	11100000 10100100 10001011 00001010

Raw vs. Curated Data



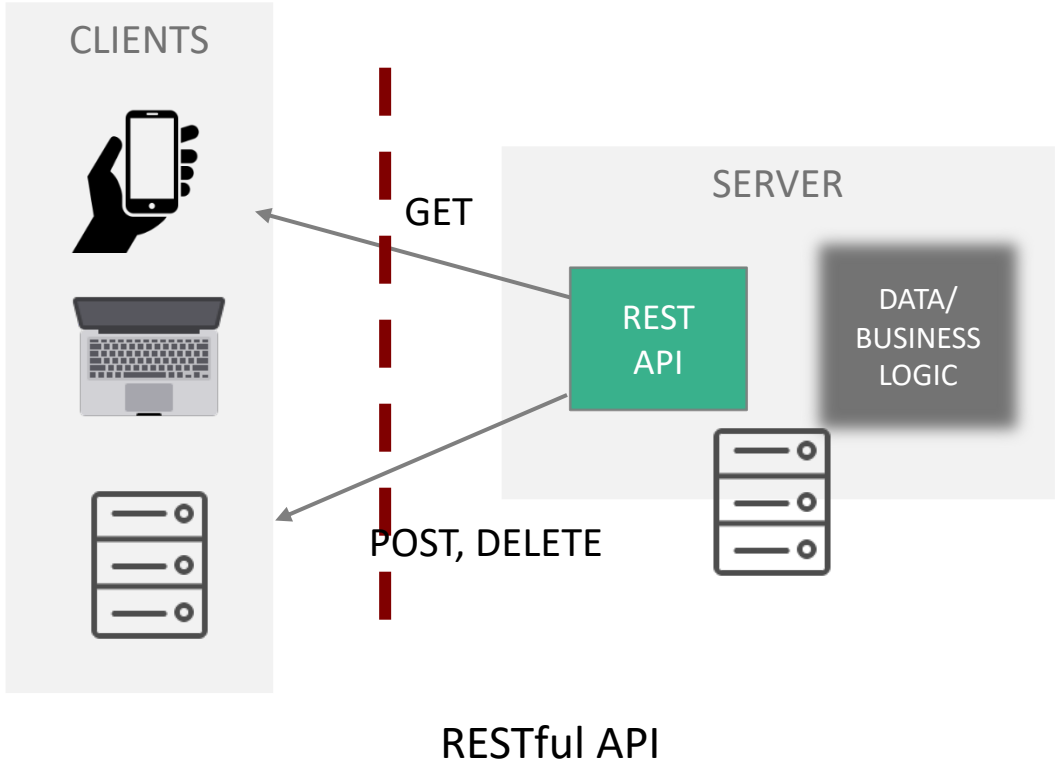
Provenance — where data came from and what happened to it — must be preserved at every stage

Never overwrite the raw copy

Keep the original response/file untouched — reprocess from raw if a downstream bug is found

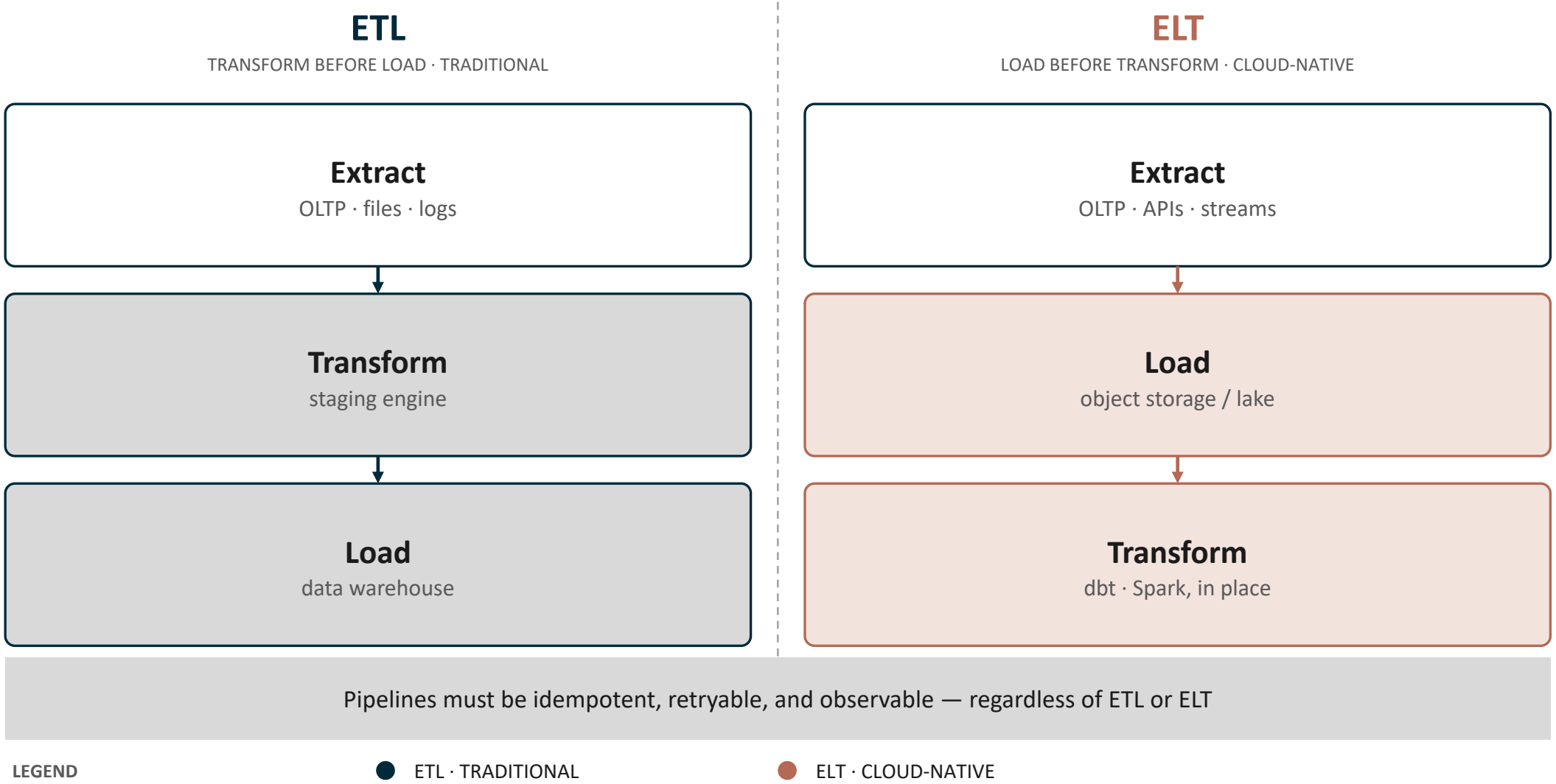
Application Programming Interfaces (API)

- API
  - A set of rules that allows different software applications to communicate.
  - Defines how requests and responses should be structured.
  - Example: A weather app uses an API to retrieve current weather data.
- Web Services
  - Services that enable communication between applications over a network, usually the Internet.
  - Common protocols include REST, SOAP, and GraphQL.
  - Typically use HTTP/HTTPS for communication.

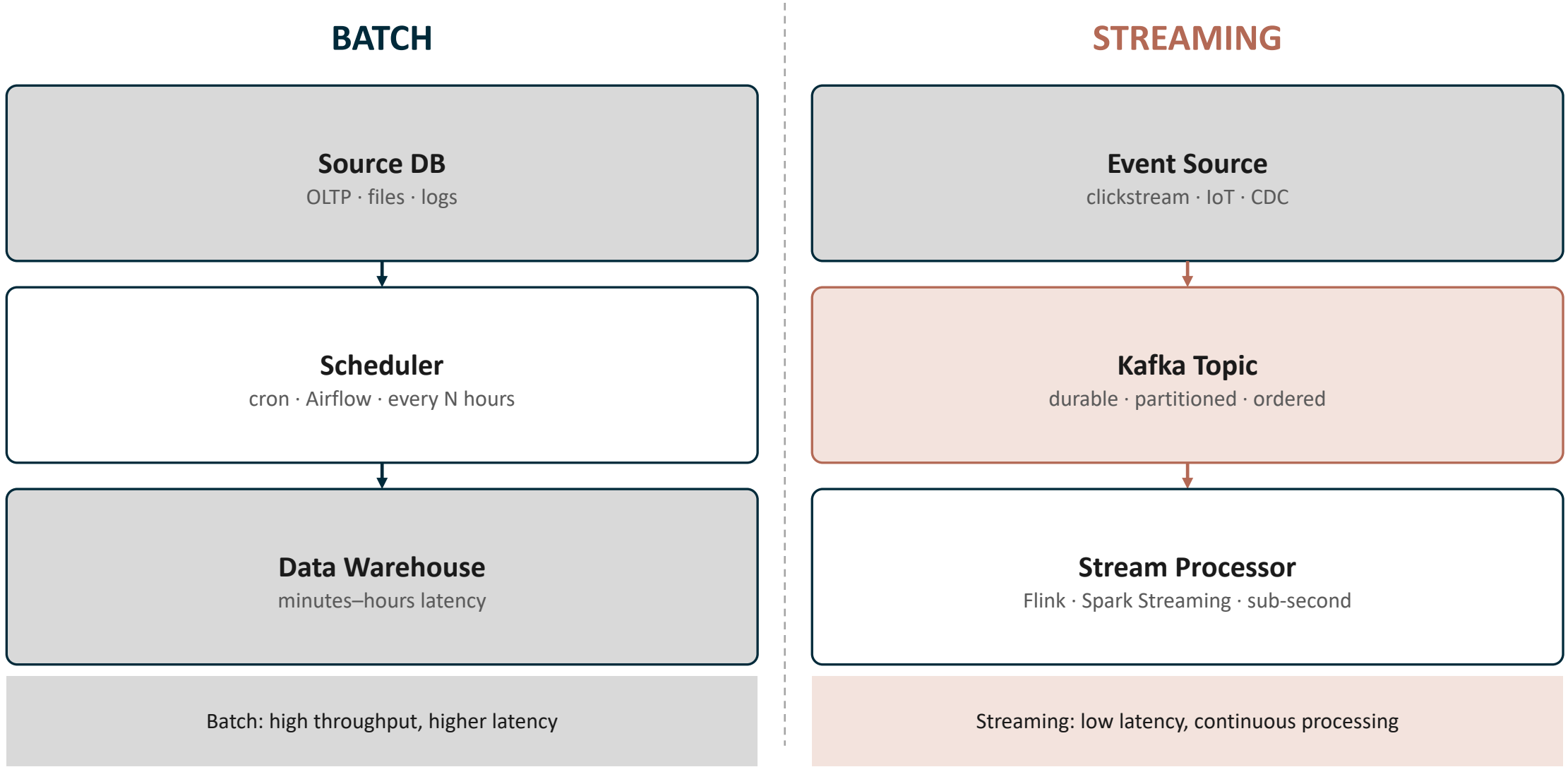


RESTful API

Data Pipeline Overview



Batch vs Streaming



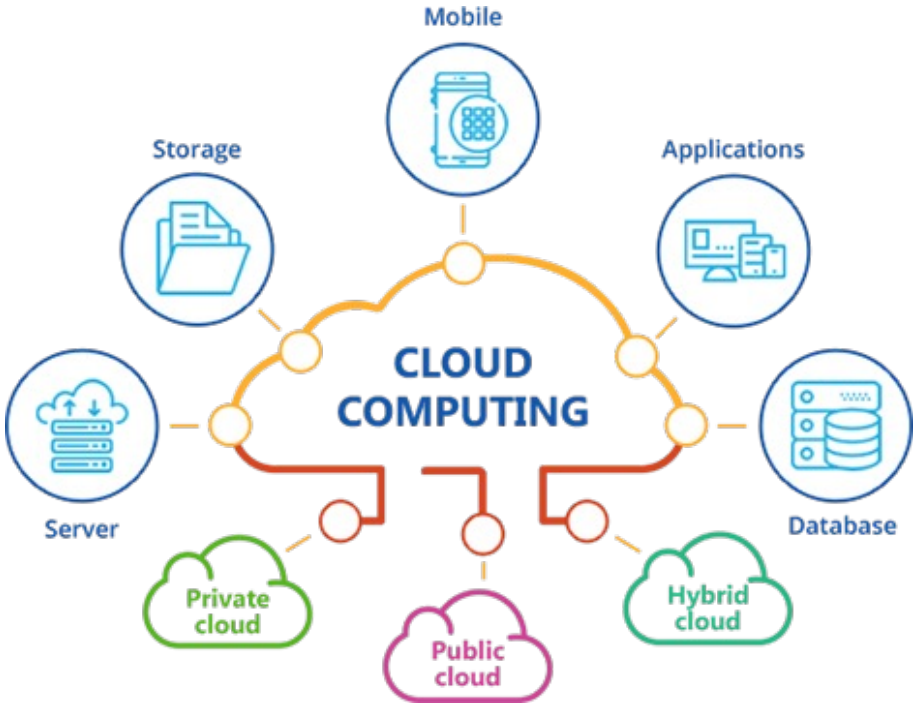


Cloud Computing Fundamentals

Module 3

Cloud Computing

- Cloud Computing is the central storage of computing resources that can be accessed on-demand from anywhere through any type of device with minimal management effort - NIST



Cloud Computing

On Demand



Self Service



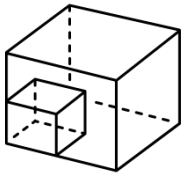
Accessibility



Utility-Based



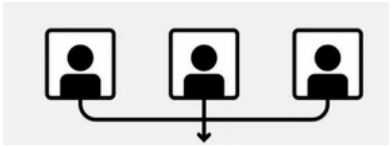
Scalable



Elastic



Multi-tenant



Infrastructure-less



Why Cloud-Native?

- On-demand elasticity: scale compute and storage independently in seconds
- Pay-per-use: no upfront hardware cost; pay for what you consume
- Managed services: cloud vendors operate the infrastructure, teams focus on data
- Global reach: deploy in any region; meet data residency requirements
- Cloud-native ≠ "lift-and-shift" — redesign for distributed, fault-tolerant systems

Cloud Computing: Benefits

- Flexibility and location independence
- Easier collaboration
- Reliability
- Increased security
- High scalability and unlimited storage capacity
- Easier and automatic software updates
- Lower maintenance and challenges with obsolete systems
- Reduced technology costs

Cloud Computing: Expenditure

CapEx

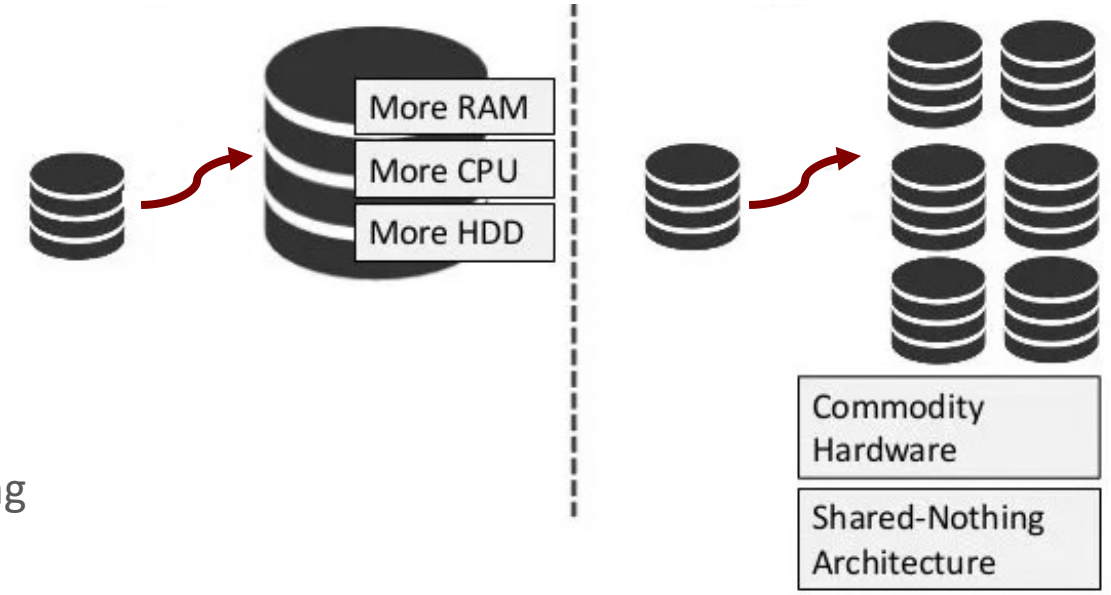
- Buying and getting stuck with capacity you don't need today to meet tomorrow's uncertainties.
- Entering vendor contracts that create business dependencies
- Long and complex acquisition process
- Paying staff to "keep the lights on"
- Letting your equipment dictate your business approach

OpEx

- Pay-as-you-go cloud services and other IT infrastructure
- Services are provided instantly and on demand without long term contracts
- Quickly and easily reconfigure capacity based on requirements
- Gives businesses agility and flexibility
- Delete assets that are not needed

Cloud Computing: Scalability

- Capability of a system, network, or process to handle a growing amount of work, or its potential to be enlarged to accommodate that growth.



Vertical Scaling
(Scale Up)

Horizontal Scaling
(Scale Out)

Infrastructure as a Service (IaaS)

- Enables users to spin up computing power, databases or other infrastructure without the labor-intensive server management or hardware investments.
- Users are responsible for installing and maintaining the operating system, databases, security components, and applications.

Google CE

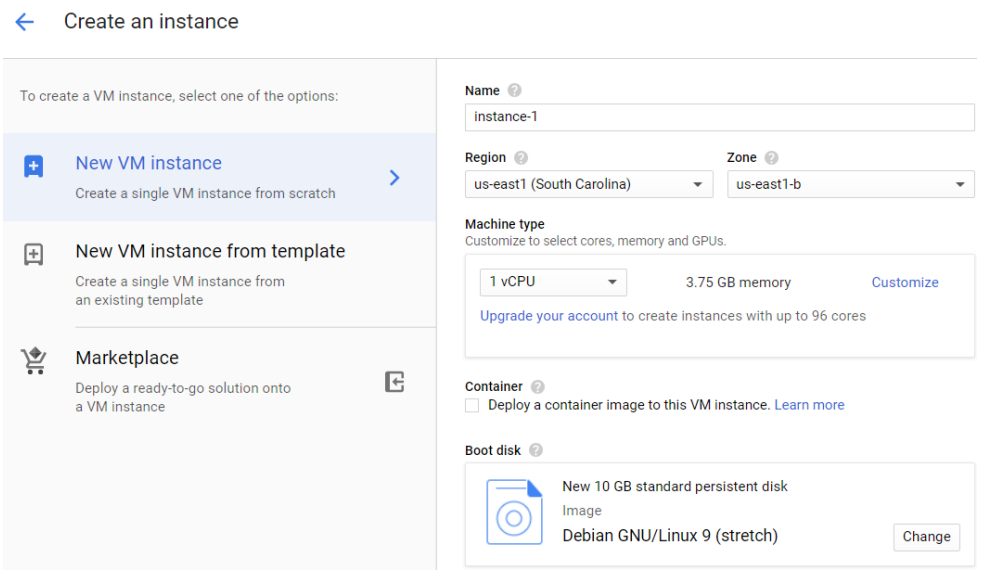


Amazon EC2



Azure VM

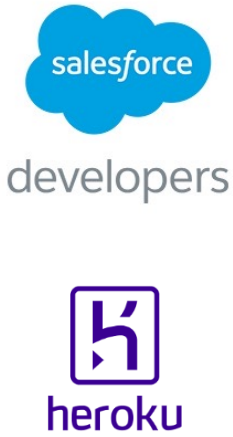




Platform as a Service (PaaS)

- PaaS includes APIs and tools that enable developers to hook in features such as traffic splitting, monitoring, and version control systems.
- Developers spend more time developing, testing, and deploying applications instead of dealing with system tasks such as managing OS, installation and patches







Software as a Service (SaaS)

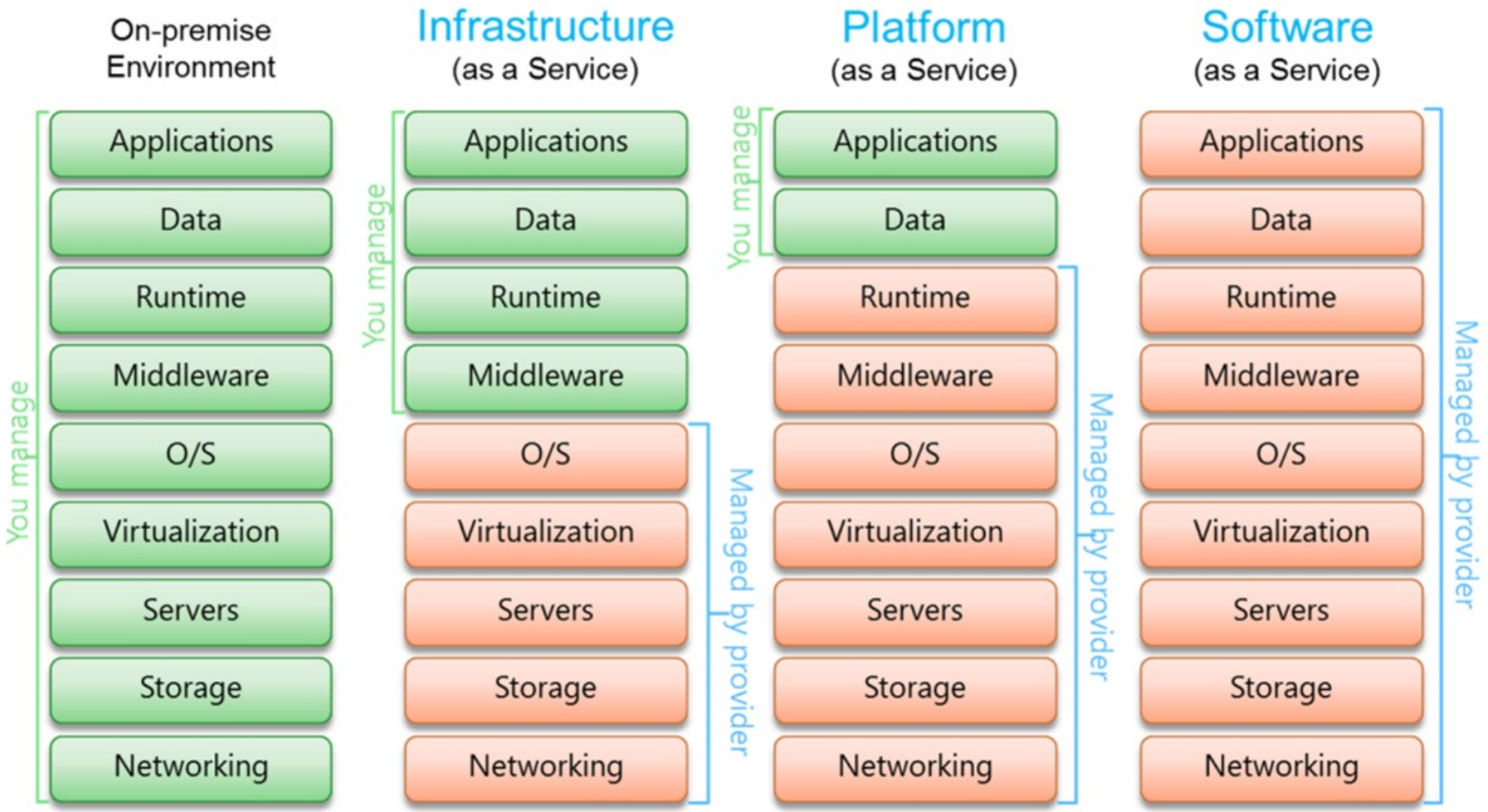
- SaaS providers host an application and make it available to users through the internet, usually a browser-based interface
- As the most familiar category of cloud computing, users most commonly interact with SaaS applications







Comparison of Cloud Services



Cloud Providers



















Regions

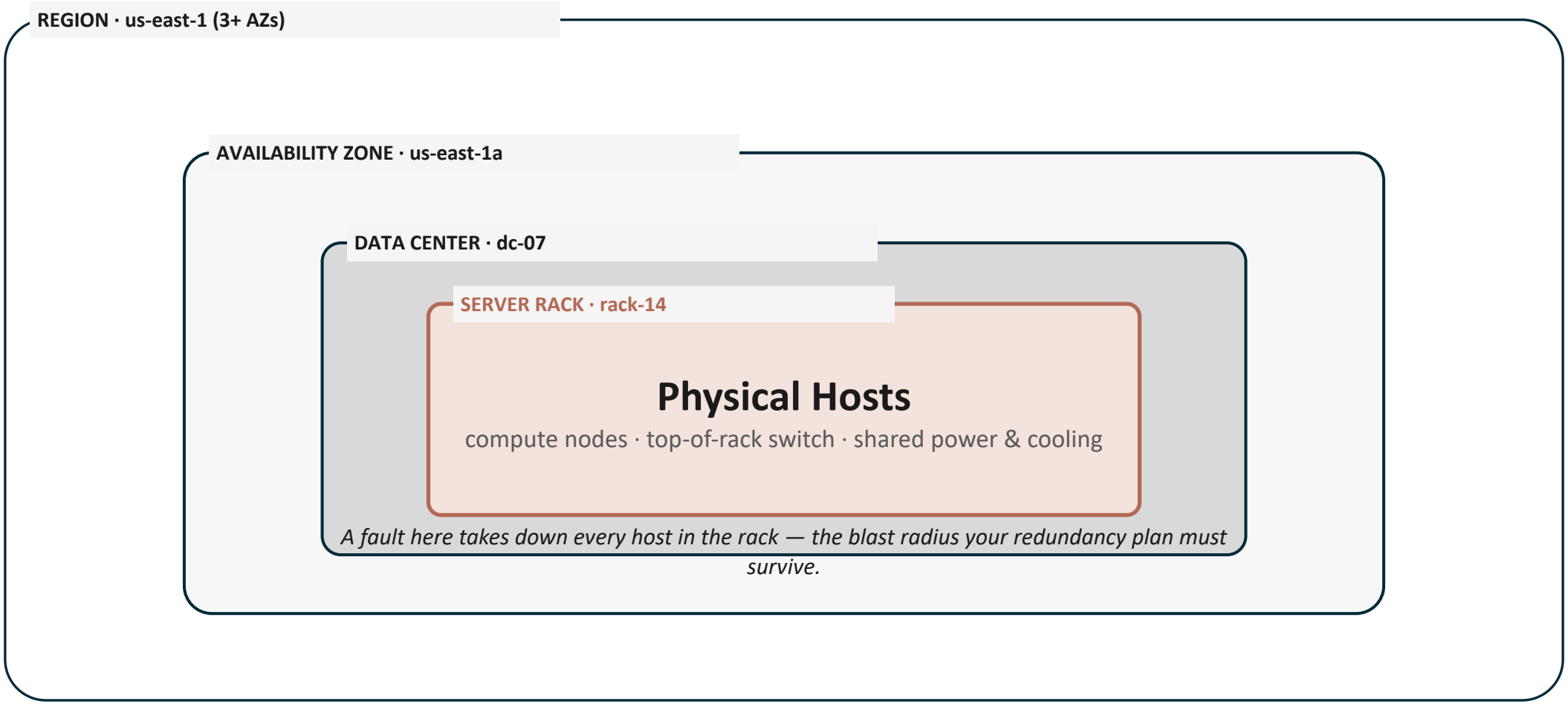


AWS



GCP

Global Infrastructure



EACH LAYER IS A DIFFERENT FAILURE DOMAIN AND A DIFFERENT SLA

Availability Zones

- Availability zones (AZs) are isolated data centers located within specific regions in which public cloud services originate and operate.



Cloud Setup

- Create cloud free-tier accounts
  - AWS
  - GCP
- SDK Installation
  - AWS Command Line Interface
  - Google Cloud CLI

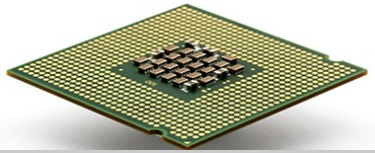


Cloud Computing Trends

Module 4

Computer Architecture







Input → Process → Output

Process → Storage

Storage → Primary Storage

Storage → Secondary Storage



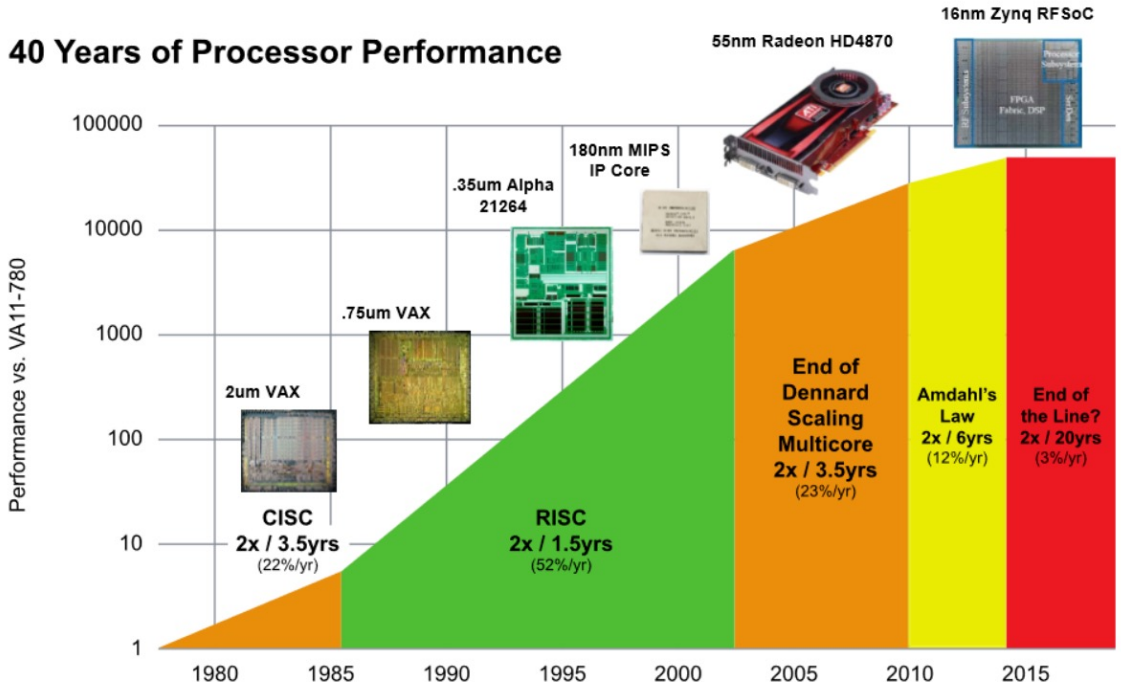


Moore’s Law

- The number of transistors in a integrated circuits doubles approximately every 2 years. This also has an impact on processing speeds and cost of computing



Gordon Earle Moore
Intel Co-founder



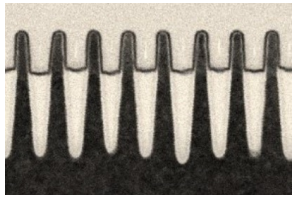
Source: John Hennessy and David Patterson, Computer Architecture: A Quantitative Approach, 6/e 2018

Moore's Law - Physical Limits

- Semiconductor Manufacturing
  - Approaching atomic limits for silicon wafer manufacturing process
- Clock Speed Limits
  - Physical limits to CPU speed due to heat generated, power consumed, gate delays, etc.
- Cost vs Benefit

Year	Node (nm)	Lithography Technology
1957	254000	Camel's Hair Brush
1958	127000	Silk Screen Printer
1959	76200	Contact Printer – Emulsion plates
1964	16000	Contact Printer – Chrome plates
1972	8000	Proximity Aligner
1974	5000	Projection Aligner
1982	2000	g-line Stepper
1990	800	i-line Stepper
1997	250	248nm Scanner
2003	90	193nm Scanner
2009	32	193nm Immersion Scanner
2019	7	EUV Scanner

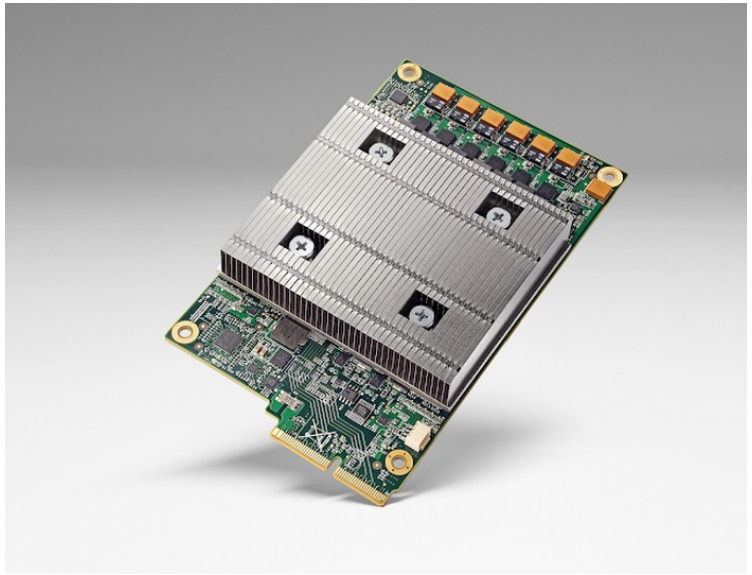
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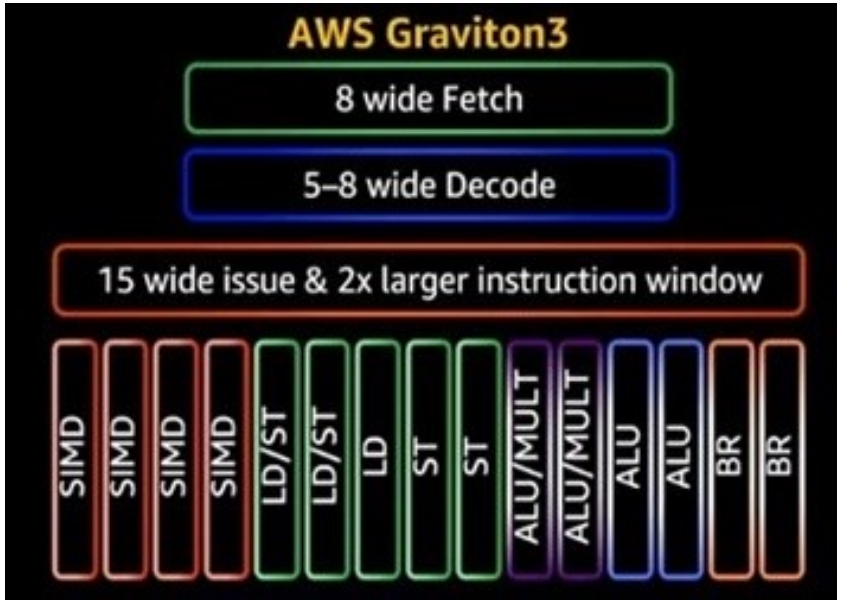
7nm transistors packed below 30nm fin pitch. Source: IBM

Cloud Microprocessors

- Cloud providers give users access to custom microprocessors and ASICS



GCP : Tensor Processing Unit (TPU)



AWS Graviton

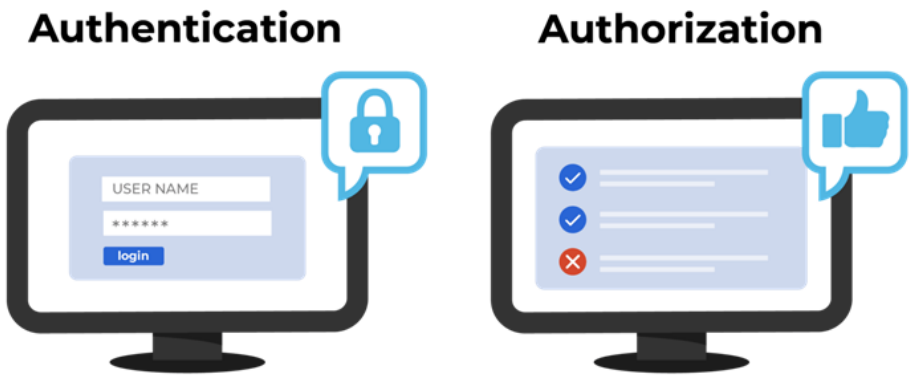
Data Governance

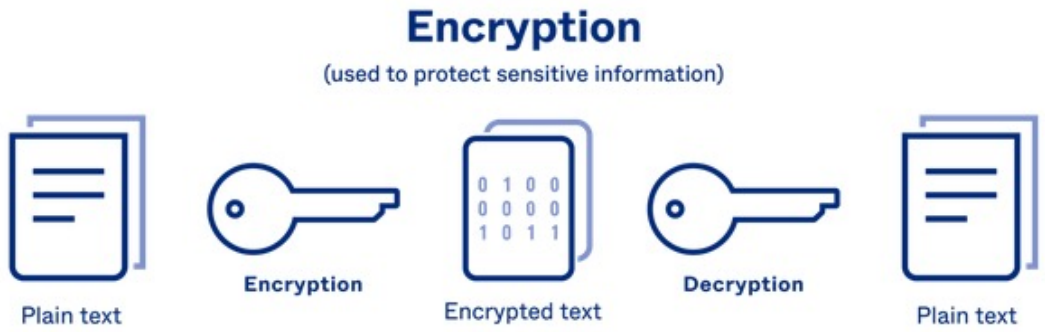
- Process of managing availability, usability, integrity, and security of data in enterprise systems.
- Exercise of authority and control over the management of data assets
- Creation and enforcement of policies and standards
- Activities ensuring compliance with data governance policies



Source: Damadach

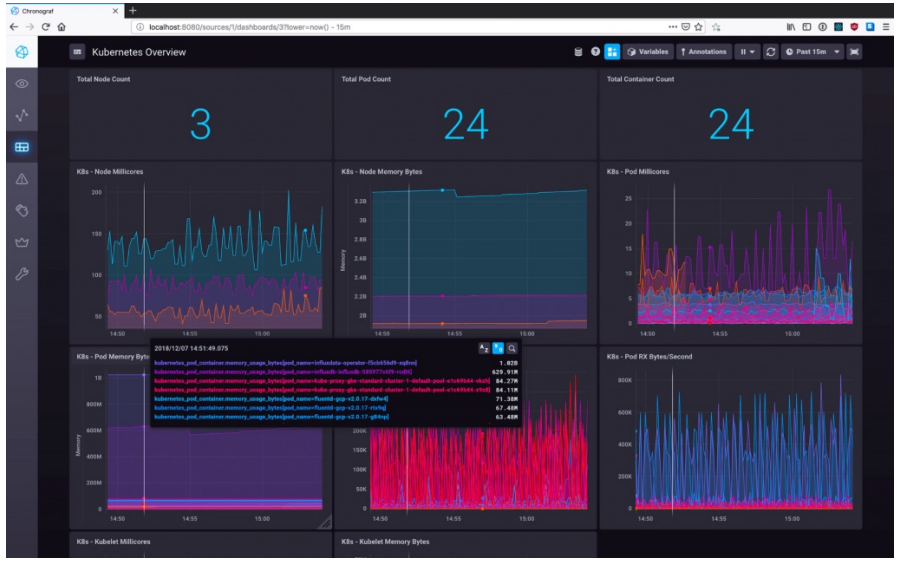
Data Security

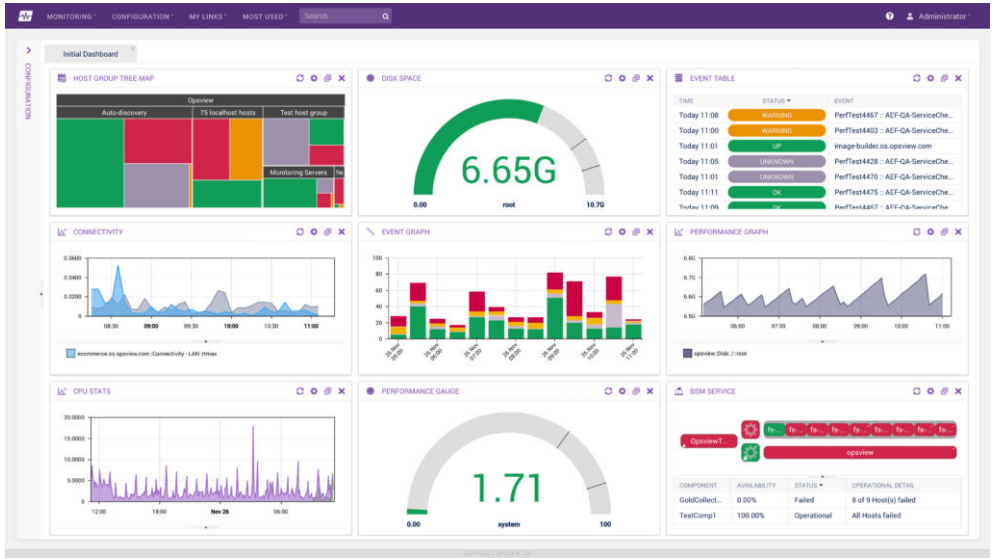




Data Monitoring

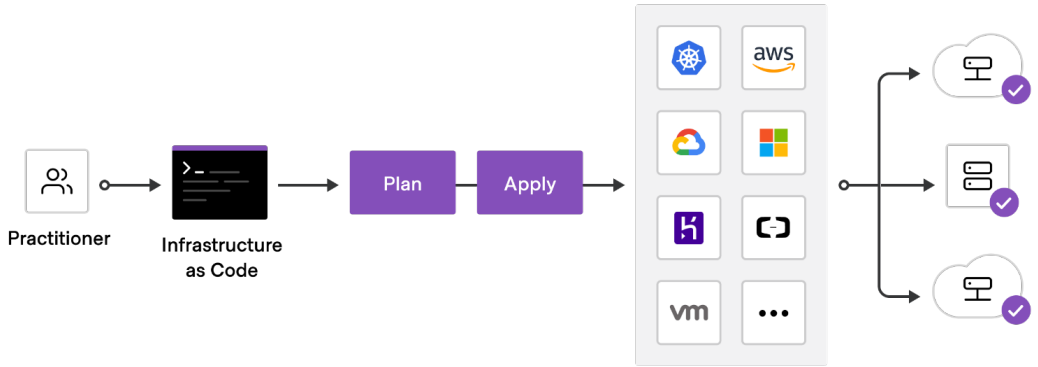
- Oversight mechanisms that monitor and ensure quality on each data instance created, utilized, and maintained within an organization





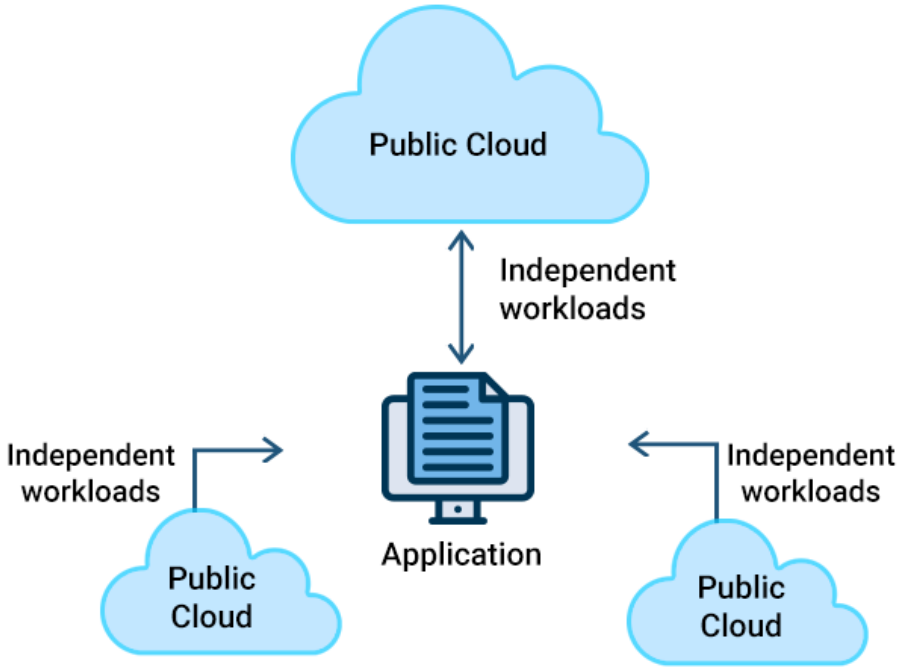
Infrastructure As Code (IaC)

- Uses DevOps methodology and versioning with a descriptive model to define and deploy infrastructure, such as networks, virtual machines, load balancers, and connection topologies
- Benefits
  - Align development and operations
  - Cost reduction
  - Increase in speed of deployments
  - Reduce errors
  - Improve infrastructure consistency
  - Eliminate configuration drift
- Popular Tools
  - Chef, Puppet, Ansible, Saltstack, Terraform, AWS CloudFormation



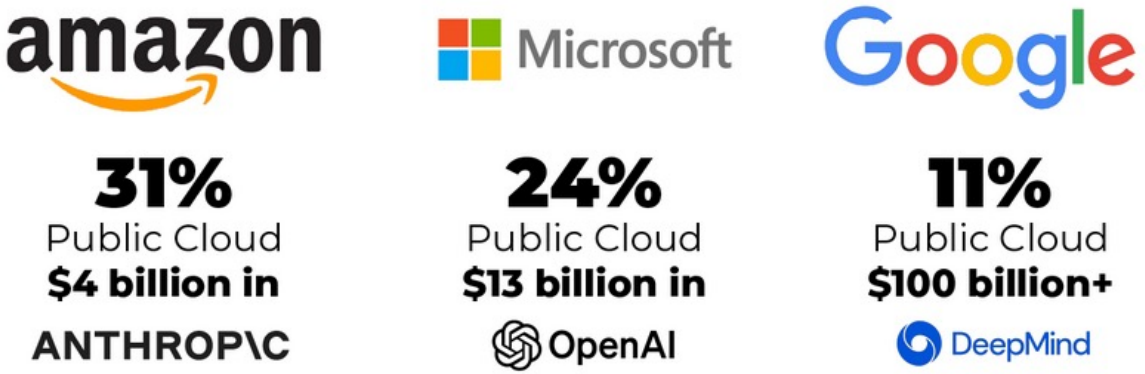
Multi-cloud Strategy

- Cloud approach made up of more than one cloud service, from more than one cloud vendor
- Benefits:
  - Flexibility
  - Proximity
  - Failover
  - Cost advantages
- Challenges:
  - Complexity
  - Managing and automating environments



Generative and Agentic AI

- Cloud is evolving into the primary delivery layer for intelligent, autonomous applications, not just compute and storage.
  - Drives new revenue streams (AI-as-a-Service)
  - Increases customer lock-in via integrated AI workflows
  - Enables enterprise automation at scale
  - Positions cloud providers as end-to-end AI platforms



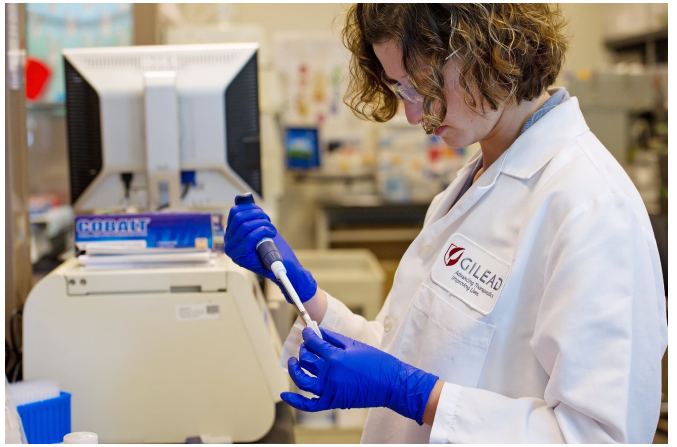


Success Stories

Module 5

Health Care and Life Sciences

- Opportunity
  - Gilead wanted to modernize its data infrastructure and use cloud innovation to improve operational performance
- Solution
  - Adopted a data mesh approach to improve agility, accelerate insight generation, and increase its return on investment.
  - The underlying architecture uses several major AWS services: Amazon S3, Amazon RDS, Amazon Redshift (Datawarehouse), SAP HANA on AWS, AWS Data Exchange, etc.
- Outcomes
  - 50 PB of data managed on the cloud
  - 70% of data center footprint migrated to cloud
  - Enhanced sustainability and automated compliance

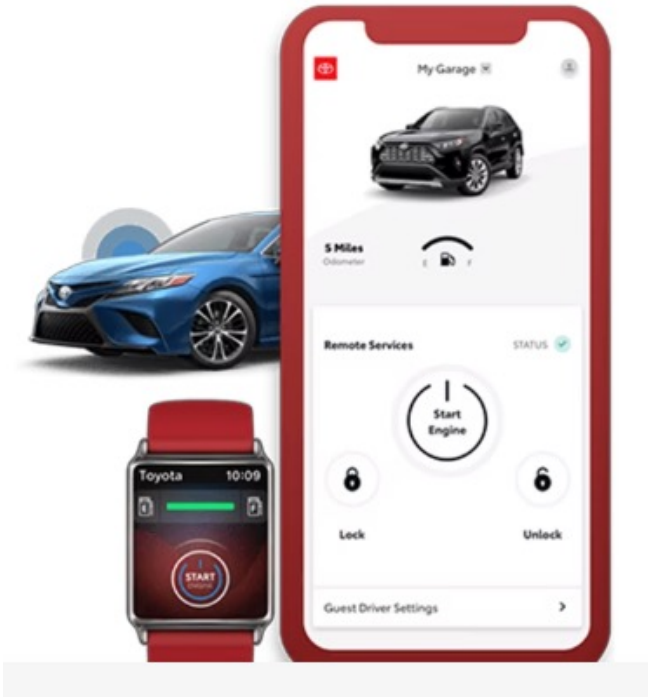


GILEAD

https://aws.amazon.com/

Automotives – Connected Cars

- Opportunity
  - Toyota Connected wanted to evolve from an Automotive to a Mobility Services Business
- Solution
  - Used AWS Lambda, a serverless compute service that enables companies to run code without provisioning or managing servers; it also uses Amazon KDS, a real-time data streaming service, to capture data and enable analytics.
- Outcomes
  - 70–80% cost-of-ownership savings
  - Built next generation mobility platform on the cloud
  - Reinventing the development of connected vehicles while reducing costs up to 80%.



TOYOTA connected

https://aws.amazon.com/

Automotives – Connected Cars

Azure

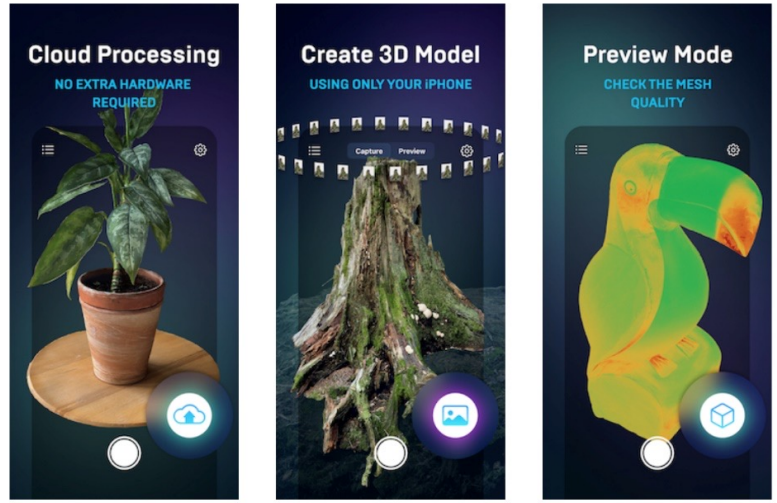
- Opportunity
  - Drivers experience enhancement
- Solution
  - Used Azure OpenAI Service to enable
- Outcomes
  - Elevated voice command and interaction
  - Expanded task capability
  - Contextual follow-up questions
  - Integration with third-party services



https://azure.microsoft.com

Gaming and Entertainment

- Opportunity
  - Build out a highly scalable next gen gaming platform
- Solution
  - Epic Games (Epic) moved to AWS in 2018 to deliver the storage, analytics, and scaling capabilities critical to its business
- Outcomes
  - Fortnite runs almost entirely on AWS, including its worldwide game-server fleet, backend services, and websites.
  - Supports over 15.3 million concurrent players during live events
  - Creators can Build Over 2 Million Digital Humans



https://aws.amazon.com/

Summary

- Data engineers design, build, and operate the pipelines that make data useful — distinct from, but collaborative with, data scientists and analysts.
- Structured, semi-structured, and unstructured data need different storage and processing approaches; most real-world data is unstructured.
- Format choice (row vs. columnar vs. open table formats) affects storage cost, query speed, and interoperability.
- Good pipelines — whether ETL or ELT — are idempotent, retriable, and observable, and never overwrite the raw copy.
- Cloud computing trades CapEx for OpEx and delivers compute, storage, and services on demand; IaaS, PaaS, and SaaS trade off control for managed convenience.
- Regions and availability zones isolate failure domains; provider and region choice